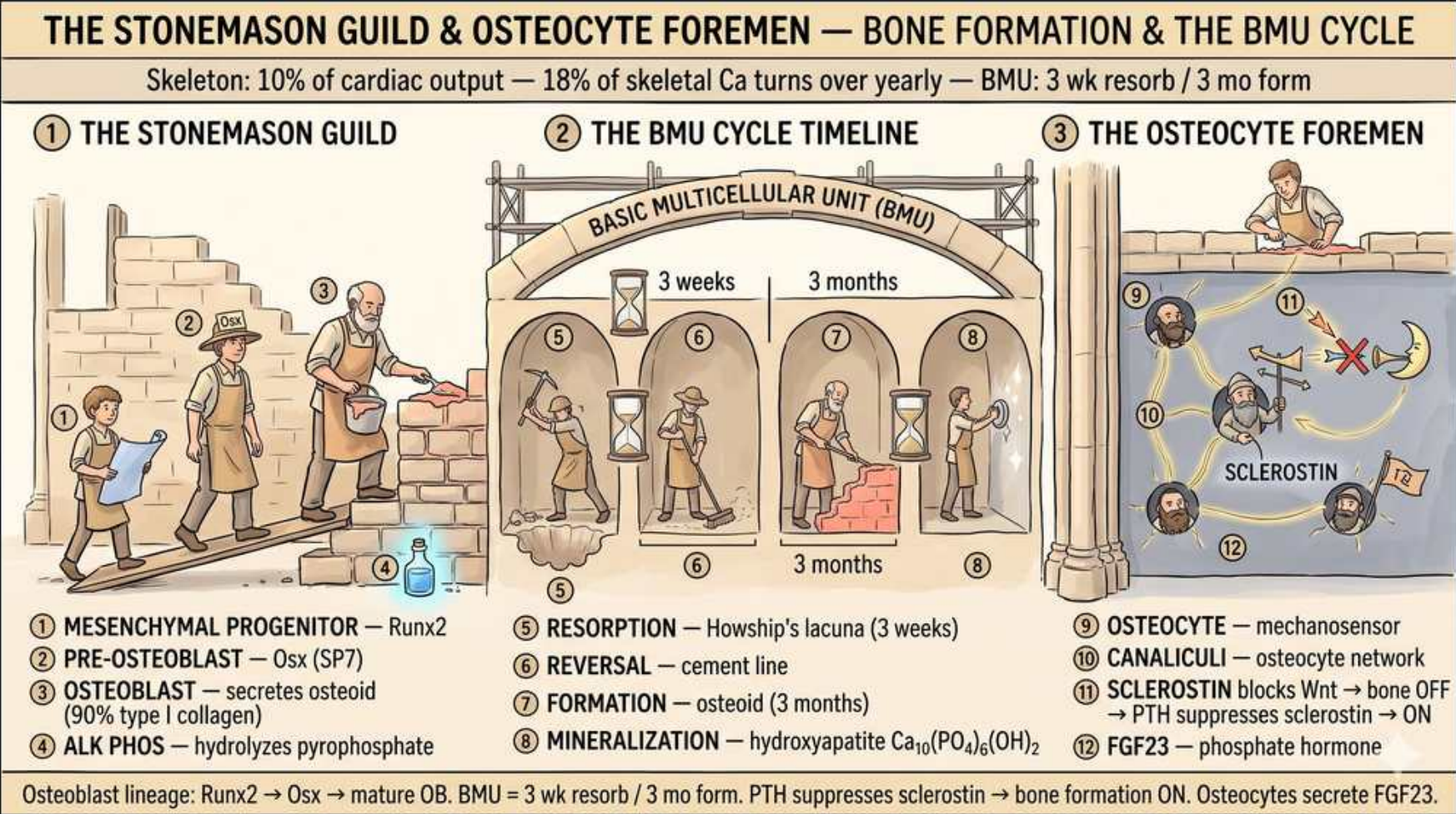




1. Skeleton turnover facts

WHAT YOU SEE

The header banner over THE STONEMASON GUILD reads 'Skeleton: 10% of cardiac output — 18% of skeletal Ca turns over yearly — BMU: 3 wk resorb / 3 mo form' — bone drawn as a busy active construction site, not inert stone.

WHAT IT ENCODES

Bone is metabolically active tissue, not inert scaffolding: it receives ~10% of cardiac output and ~18% of skeletal calcium is exchanged yearly. The basic multicellular unit (BMU) is the team of osteoclasts and osteoblasts that continuously remodels bone in a tightly coupled cycle. ~10% of the adult skeleton is replaced per year; trabecular (cancellous) bone turns over faster than cortical bone, which is why steroid-induced and high-turnover bone loss hits the spine and hip first. Remodeling balances mechanical repair, calcium homeostasis, and phosphate handling.

BOARD TRAP

Board trap: a question may imply bone is a static calcium reservoir that only releases Ca when blood is low. WRONG — bone is constantly remodeled even at normal serum Ca. The discriminator: serum calcium is held in a tight range (8.5–10.5 mg/dL) by PTH/vitamin D regulating intestinal absorption, renal reabsorption AND bone flux simultaneously, not by passive leaching.

2. Mesenchymal progenitor (Runx2)

WHAT YOU SEE

The youngest apprentice mason (figure #1) reading blueprints, tagged MESENCHYMAL PROGENITOR — Runx2 — the rookie laying out plans is the Runx2-committed stem cell that starts the osteoblast lineage.

WHAT IT ENCODES

The osteoblast lineage begins with a mesenchymal stem cell (MSC) committed by the master transcription factor Runx2 (CBFA1). Runx2 is absolutely required — Runx2-knockout produces a cartilaginous skeleton with no mineralized bone. The same MSC pool can become osteoblasts, chondrocytes, or adipocytes; aging/steroids shunt MSCs toward fat (marrow adiposity), reducing bone formation. Wnt/ β -catenin signaling pushes the MSC toward bone and away from fat.

BOARD TRAP

Board trap: stem says osteoclasts and osteoblasts share a precursor. WRONG. Discriminator: OsteoBLASTs are MESENCHYMAL (Runx2 → Osterix); osteoCLASTs are HEMATOPOIETIC, arising from the monocyte/macrophage line under M-CSF and RANKL. They meet only at the BMU. Mixing up the two lineages is a classic distractor.

3. Pre-osteoblast (Osx/SP7)

WHAT YOU SEE

The middle mason (#2) wearing an 'Osx' cap, tagged PRE-OSTEOBLAST — Osx (SP7) — the in-training mason marked Osx is the pre-osteoblast committed downstream of Runx2.

WHAT IT ENCODES

Osterix (Osx, gene SP7) acts strictly DOWNSTREAM of Runx2 to commit the pre-osteoblast to a mature, matrix-secreting osteoblast. Runx2 turns a mesenchymal cell into a pre-osteoblast; Osterix finishes the job. Wnt/LRP5-6 co-receptor signaling and BMP-2 reinforce this step. Loss-of-function LRP5 → osteoporosis-pseudoglioma; gain-of-function LRP5 → high bone mass.

BOARD TRAP

Board trap: a sequence question reverses the order, placing Osterix upstream of Runx2. WRONG — the fixed order is Runx2 → Osterix → osteoblast. Discriminator: Runx2-null animals make no bone at all, whereas Osterix-null still form pre-osteoblasts, proving Runx2 comes first.

4. Osteoblast (osteoid)

WHAT YOU SEE

The senior bearded mason (#3) troweling fresh red mortar onto the wall, tagged OSTEOBLAST — secretes osteoid (90% type I collagen) — laying mortar = the osteoblast secreting collagen-rich osteoid.

WHAT IT ENCODES

Mature osteoblasts secrete osteoid, the unmineralized organic matrix that is ~90% type I collagen plus osteocalcin, osteopontin, and osteonectin. Osteoid is laid down first and mineralized ~10 days later; the gap between osteoid and the mineralization front is the osteoid seam, which is widened in osteomalacia. Osteoblasts also produce RANKL and OPG, thereby controlling osteoclast activity — they are the command center of remodeling. Osteocalcin is a serum bone-formation marker (vitamin K– and vitamin D–dependent).

BOARD TRAP

Board trap: stem attributes type I collagen defects to osteomalacia. WRONG. Discriminator: defective TYPE I COLLAGEN = osteogenesis imperfecta (blue sclerae, fractures, COL1A1/COL1A2); osteomalacia is normal collagen that fails to MINERALIZE (vitamin D/phosphate problem). Different osteoid, different disease.

5. Alkaline phosphatase

WHAT YOU SEE

A blue bottle at the masons' feet (#4) labelled ALK PHOS — hydrolyzes pyrophosphate — the bottle of solvent that dissolves the pyrophosphate inhibitor so mineral can set.

WHAT IT ENCODES

Osteoblast tissue-nonspecific alkaline phosphatase (ALP) hydrolyzes pyrophosphate, an endogenous mineralization INHIBITOR, freeing inorganic phosphate locally so hydroxyapatite can deposit. ALP is a serum bone-FORMATION/turnover marker. It rises in Paget disease (often markedly, hundreds), osteomalacia, healing fractures, bony metastases, and normal adolescent growth. Hypophosphatasia (loss-of-function ALP) causes accumulated pyrophosphate and defective mineralization.

BOARD TRAP

Board trap: an elderly patient with isolated very high ALP, normal Ca/phosphate, normal GGT — distractor calls it cholestatic liver disease. WRONG. Discriminator: check GGT or fractionate ALP — a NORMAL GGT with high ALP points to BONE origin (think Paget), not liver. In primary hyperparathyroidism ALP is normal-to-mildly elevated, not in the hundreds.

6. Resorption (3 weeks)

WHAT YOU SEE

Under an hourglass marked '3 weeks', a mason (#5) swinging a PICKAXE to dig out the wall, tagged RESORPTION — Howship's lacuna — pickaxe demolition = osteoclast resorption, the fast 3-week phase.

WHAT IT ENCODES

The BMU cycle opens with osteoCLASTic RESORPTION lasting ~3 weeks. Multinucleated osteoclasts dock onto bone via $\alpha v \beta 3$ integrin, form a sealing zone and ruffled border, then secrete H⁺ (via carbonic anhydrase II and a vacuolar H⁺-ATPase) to dissolve hydroxyapatite, plus cathepsin K to digest collagen — excavating Howship's lacunae. This step is driven by RANKL (from osteoblasts/osteocytes); denosumab (anti-RANKL) and bisphosphonates block it. Cathepsin-K deficiency = pycnodysostosis.

BOARD TRAP

Board trap: stem lists the antiresorptive that 'turns off' osteoclasts AND builds bone in the same breath. WRONG to call denosumab/bisphosphonates anabolic. Discriminator: these are ANTIRESORPTIVE (slow breakdown); only teriparatide/abaloparatide (PTH analogs) and romosozumab (early phase) are ANABOLIC. Also: stopping denosumab causes rebound resorption and vertebral fractures — never stop without transitioning to a bisphosphonate.

7. Reversal (cement line)

WHAT YOU SEE

A worker (#6) sweeping/cleaning the excavated pit, tagged REVERSAL — cement line — clearing the rubble and laying the boundary line couples resorption to the coming formation.

WHAT IT ENCODES

The REVERSAL phase couples resorption to formation: osteoclasts undergo apoptosis and reversal cells/macrophages clean the lacuna, then deposit a thin collagen-poor cement (reversal) line that is the histologic boundary between old and new bone. Coupling signals (TGF- β , IGF-1 released from resorbed matrix, plus osteoclast-derived factors) recruit osteoblasts. The cement line is a basophilic cement seam visible on histology and a hint at how recently a unit remodeled.

BOARD TRAP

Board trap: question implies resorption and formation happen independently/simultaneously at random sites. WRONG. Discriminator: in healthy bone they are COUPLED and sequential at the same BMU — resorption first, then formation. Uncoupling (e.g., glucocorticoids suppress formation, hyperthyroidism/hyperPTH accelerate resorption) is precisely what causes net bone loss.

8. Formation (3 months)

WHAT YOU SEE

Under an hourglass marked '3 months', a mason (#7) laying new red bricks back into the cavity, tagged FORMATION — osteoid — slow bricklaying = osteoblastic refill, four times longer than resorption.

WHAT IT ENCODES

FORMATION is the slow phase: osteoblasts refill the resorption cavity with osteoid over ~3 MONTHS — roughly four times longer than the ~3-week resorption phase. This asymmetry means anything that increases remodeling frequency (high turnover) produces a transient bone deficit and increased fracture risk, because many cavities are open and unfilled at any moment. Antiresorptives reduce the number of remodeling sites; anabolics increase osteoblast number/activity to fill faster.

BOARD TRAP

Board trap: stem claims antiresorptives work by speeding bone formation. WRONG. Discriminator: bisphosphonates/denosumab work by REDUCING resorption AND the number of active BMUs, allowing existing cavities to fill and secondary mineralization to mature — DXA gains are mostly from filling the remodeling space, not new modeling. The slow 3-month fill is also why DXA response takes 1–2 years.

9. Mineralization (hydroxyapatite)

WHAT YOU SEE

The final mason (#8) finishing/hardening the rebuilt wall, tagged MINERALIZATION — hydroxyapatite $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$ — the cured solid wall is osteoid hardened into mineral.

WHAT IT ENCODES

Osteoid is mineralized by deposition of hydroxyapatite $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$, requiring adequate calcium, phosphate, and alkaline phosphatase, plus 1,25-dihydroxyvitamin D to keep the $\text{Ca} \times \text{PO}_4$ product sufficient. Primary mineralization occurs within days at the osteoid seam; secondary mineralization (crystal maturation) continues for months, increasing bone density. Failure of this step with continued osteoid production = OSTEOMALACIA in adults / RICKETS in children (widened osteoid seams, Looser zones/pseudofractures).

BOARD TRAP

Board trap: a patient with bone pain, low-normal Ca, LOW phosphate, HIGH ALP, HIGH PTH — distractor calls it primary hyperparathyroidism. WRONG. Discriminator: that pattern (low/normal Ca + low PO_4 + high ALP + SECONDARY high PTH + low 25-OH vitamin D) is OSTEOMALACIA from vitamin D deficiency. Primary hyperPTH has HIGH calcium with inappropriately high PTH.

10. Osteocyte (mechanosensor)

WHAT YOU SEE

Inside THE OSTEOCYTE FOREMEN arch, a bearded foreman face walled into the stone (#9), tagged OSTEOCYTE — mechanosensor — the mason entombed in the wall who now supervises is the load-sensing osteocyte.

WHAT IT ENCODES

Osteocytes are terminally differentiated osteoblasts entombed in lacunae within mineralized matrix; they are the most abundant bone cell (>90%) and the master regulators of remodeling. They sense mechanical load (and microdamage) via fluid shear through canaliculi and translate it into signals — increasing sclerostin when unloaded (bone OFF) and decreasing it when loaded (bone ON). They also secrete FGF23 (phosphate control) and RANKL, directing both osteoclasts and osteoblasts.

BOARD TRAP

Board trap: stem credits osteoBLASTs as the principal mechanosensors that detect strain. WRONG. Discriminator: the embedded OSTEOCYTE is the mechanosensor and the chief source of sclerostin and FGF23. This is also why immobilization/microgravity (mechanical unloading) raises sclerostin and causes disuse osteoporosis and hypercalciuria.

11. Canaliculi (osteocyte network)

WHAT YOU SEE

Glowing threads linking the entombed foreman faces (#10), tagged CANALICULI — osteocyte network — the connecting channels are the canalicular network wiring osteocytes together.

WHAT IT ENCODES

Canaliculi are the microscopic channels that connect osteocyte lacunae into a vast syncytial network, with osteocyte dendritic processes joined by gap junctions (connexin-43). This lacuno-canalicular system carries interstitial fluid whose shear stress, generated by mechanical loading, is how osteocytes detect strain and microcracks. The network lets osteocytes communicate load and damage signals to surface osteoblasts/lining cells and rapidly exchange mineral with the circulation (osteocytic osteolysis under high PTH).

BOARD TRAP

Board trap: a stem treats osteocytes as inert, isolated cells walled off from circulation. WRONG. Discriminator: via the canalicular network osteocytes are metabolically active and connected — they perform rapid mineral exchange and, under sustained high PTH, osteocytic osteolysis, contributing to the hypercalcemia of hyperparathyroidism beyond classic osteoclastic resorption.

12. Sclerostin / Wnt

WHAT YOU SEE

A foreman holding a STOP flag with a red X over the moon, tagged SCLEROSTIN — blocks Wnt → bone OFF; PTH suppresses sclerostin → ON — the STOP-flag worker is sclerostin shutting bone formation off.

WHAT IT ENCODES

Sclerostin, the product of the SOST gene secreted by osteocytes, binds LRP5/6 to INHIBIT Wnt/ β -catenin signaling, switching bone formation OFF. Mechanical loading and intermittent PTH SUPPRESS sclerostin → Wnt ON → osteoblast activity. Loss of sclerostin (SOST mutations) causes sclerosteosis and van Buchem disease (massive high bone mass). The therapeutic exploit is romosozumab, a monoclonal anti-sclerostin antibody.

BOARD TRAP

Board trap: stem offers romosozumab to a patient with recent MI/stroke as first-line for osteoporosis. WRONG choice. Discriminator: romosozumab carries a BOXED WARNING for myocardial infarction, stroke, and cardiovascular death — avoid within 1 year of MI/stroke. It is reserved for very-high-risk osteoporosis and is given for only 12 months (anabolic window), then followed by an antiresorptive.

13. FGF23 (phosphate)

WHAT YOU SEE

A foreman face labelled FGF23 lower in the arch (#12), tagged FGF23 — phosphate hormone — the osteocyte-foreman who fires off FGF23 to control phosphate.

WHAT IT ENCODES

FGF23 is the phosphate-regulating hormone secreted by osteocytes/osteoblasts; with co-receptor Klotho it acts on the kidney to (1) DOWNREGULATE the NaPi-2a/2c cotransporter → phosphaturia and (2) SUPPRESS 1-alpha-hydroxylase → lower 1,25-vitamin D. Net effect: it LOWERS serum phosphate. It rises early in CKD as the first defense against phosphate retention. Excess FGF23 causes hypophosphatemic rickets/osteomalacia (X-linked hypophosphatemia from PHEX mutations; tumor-induced osteomalacia from mesenchymal tumors).

BOARD TRAP

Board trap: a young adult with bone pain, LOW phosphate, inappropriately LOW/normal 1,25-vitamin D, and renal phosphate wasting — distractor calls it vitamin D–deficient osteomalacia. WRONG. Discriminator: classic vitamin D deficiency raises 1,25-D and causes secondary hyperPTH, whereas FGF23 excess gives phosphaturia with LOW 1,25-D and normal PTH/Ca — think tumor-induced osteomalacia or XLH; treat the latter with burosumab (anti-FGF23).

14. PTH suppresses sclerostin

WHAT YOU SEE

The bottom banner reads 'PTH suppresses sclerostin → bone formation ON. Osteocytes secrete FGF23.' — the summary line that intermittent PTH switches bone-building ON by silencing sclerostin.

WHAT IT ENCODES

INTERMITTENT PTH exposure is bone-ANABOLIC, partly by transiently suppressing sclerostin (Wnt ON) and increasing osteoblast number/survival — the basis of teriparatide (recombinant PTH 1-34) and abaloparatide (PTHrP analog), both daily subcutaneous injections that build trabecular bone. By contrast, CONTINUOUS PTH elevation (primary or secondary hyperparathyroidism, or once-daily levels sustained too long) is net CATABOLIC, driving RANKL-mediated resorption. Anabolic agents are capped at ~2 years of lifetime use.

BOARD TRAP

Board trap: stem assumes PTH is uniformly bad for bone, or gives teriparatide to a patient with Paget disease, prior skeletal radiation, bone metastases, or hypercalcemia. WRONG. Discriminator: the key is the PULSE — INTERMITTENT PTH builds bone, CONTINUOUS high PTH destroys it. Teriparatide/abaloparatide are CONTRAINDICATED where osteosarcoma risk is increased (Paget, open epiphyses, prior radiation) and in hypercalcemia.